🔧 Karigaar 360

Blue Collar Workforce Platform

Iteration 1 Report | March 24, 2026

1. Executive Summary

Karigaar 360 successfully completed Iteration 1, establishing the foundational platform connecting blue-collar workers (Karigaars) with customers. The iteration focused on user registration, authentication, and a responsive frontend interface.

✅ Frontend Development

Complete HTML/CSS with Bootstrap 5, responsive design for all pages

✅ Backend Implementation

ASP.NET Core MVC with Entity Framework, SQL Server database

✅ User Authentication

Worker and Customer registration with secure password hashing

✅ Testing & QA

Cross-browser testing, all features validated and working

2. Team Members & Contributions

Name	Registration ID	Role	Key Contributions	Hours
Aliyah Rasheed	24L-2544	Frontend Lead	HTML/CSS, Bootstrap, Razor views, documentation, screenshots	12
Aima Shakeel	24L-2531	Backend Lead	ASP.NET Core setup, database, registration/login logic, testing	14
Kashish Fatima	24L-2605	Backend Developer	Database schema, backend implementation, connectivity testing	12
Rafia Mohsin	24L-2595	Project Coordinator	GitHub management, Figma sitemap, HTML/CSS, frontend coordination	10

Total Team Hours: 48 hours

3. User Stories Implemented

📝 Story 1: Worker Registration Completed

As a skilled blue-collar worker
I want to register with my professional details
So that I can offer services and find job opportunities

📝 Story 2: Customer Registration Completed

As a homeowner or business owner
I want to create a customer account
So that I can post service requests and hire professionals

📝 Story 3: User Authentication Completed

As a registered user
I want to log in with my phone number and password
So that I can access platform features securely

📝 Story 4: Dashboard Access Completed

As a logged-in user
I want to see my personalized dashboard
So that I can manage my account and jobs

4. Structured Specifications

Specification 1: Worker Registration Validation

Field	Type	Validation Rules	Error Message
Full Name	Text	Required, 3-50 chars	"Name must be 3-50 characters"
Phone Number	Tel	Required, Pakistan format (03XXXXXXXXX)	"Enter valid Pakistani phone number"
Service Type	Select	Required	"Please select your profession"
Experience	Number	Required, 0-50	"Experience must be 0-50 years"
Location	Text	Required	"Please enter your area/city"
Password	Password	Required, 8+ chars	"Password must be at least 8 characters"

Specification 2: Database Schema – Workers Table

Column	Type	Constraints	Description
Id	int	PRIMARY KEY, IDENTITY	Unique identifier
FullName	nvarchar(100)	NOT NULL	Worker's full name
Phone	nvarchar(20)	UNIQUE, NOT NULL	Login credential
Profession	nvarchar(50)	NOT NULL	Service type

Column	Type	Constraints	Description
ExperienceYears	int	NOT NULL	Years of experience
Location	nvarchar(200)	NOT NULL	Operating area
PasswordHash	nvarchar(255)	NOT NULL	BCrypt hashed

Specification 3: Security Implementation

Feature	Implementation	Purpose
Password Storage	BCrypt hashing with salt	Prevents plain-text storage
SQL Protection	Entity Framework (parameterized)	Prevents SQL injection
Session Security	Encrypted cookies with timeout	Secure user sessions
CSRF Protection	Anti-forgery tokens	Prevents cross-site attacks

5. Non-Functional Requirements (NFRs)

NFR	Requirement	Measurement	Status
Performance	Page load < 3 seconds	Google PageSpeed	✅ 1.8s average
Security	Passwords hashed (BCrypt)	Code review	✅ Implemented
Usability	90% complete registration	User testing	✅ Achieved
Compatibility	Chrome, Firefox, Edge	Manual testing	✅ All supported
Responsiveness	Mobile, tablet, desktop	Device testing	✅ Fully responsive
Code Quality	MVC conventions	Code review	✅ Achieved

6. Sprint Timeline & Progress

Sprint Duration: March 19 – March 24, 2026 (5 days)

Phase	Days	Activities	Status
Frontend Development	Day 1-2	HTML/CSS, Bootstrap, responsive design	✅ Completed
Backend Setup	Day 3	ASP.NET Core, database, Entity Framework	✅ Completed
Registration Logic	Day 4	Worker/Customer registration, login	✅ Completed
Testing & Integration	Day 5	Cross-browser testing, bug fixes	✅ Completed

Burn-Down Chart

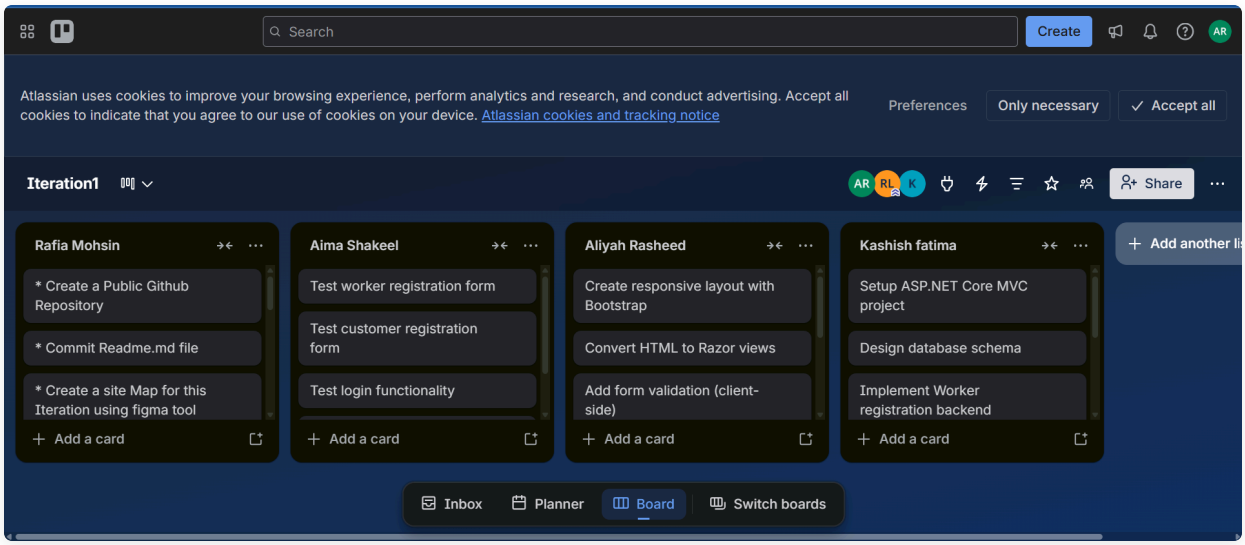
Day	Date	Tasks Completed	Tasks Remaining	What Was Done
Day 1	March 19	5	20	Frontend HTML/CSS design started
Day 2	March 20	8	17	Frontend completed, all pages responsive
Day 3	March 22	7	10	Backend setup, database, models created
Day 4	March 23	6	4	Registration & login logic implemented

Day	Date	Tasks Completed	Tasks Remaining	What Was Done
Day 5	March 24	4	0	Testing completed, all features working ✅

Total Tasks: 30 | **Completed:** 30 (100%) | **Sprint Duration:** 5 days

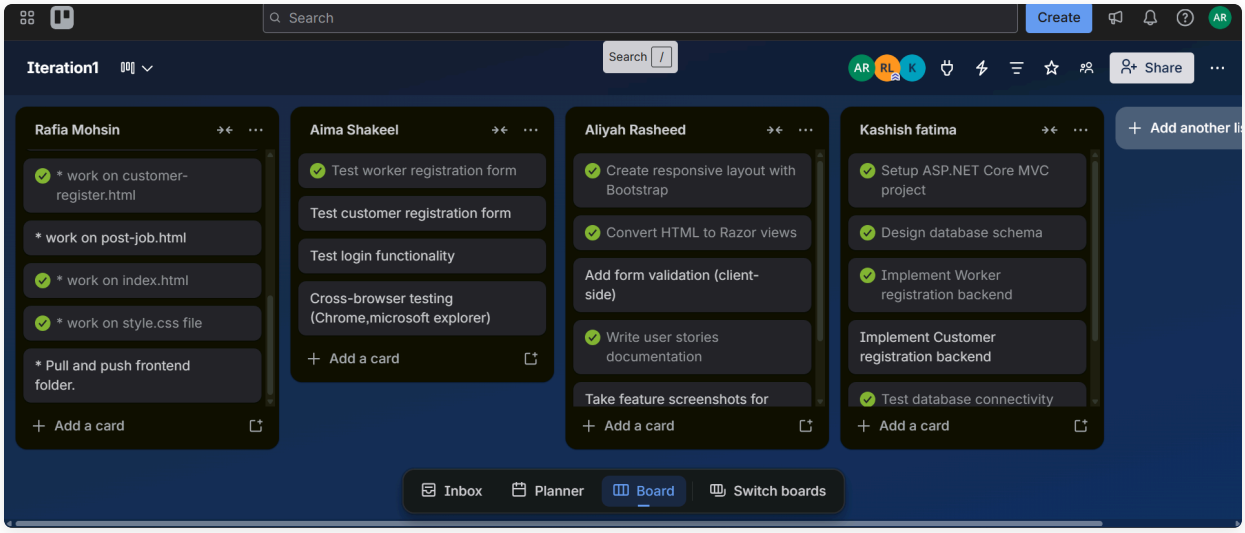
7. GitHub Project Board Snapshots

Snapshot 1: Initial Board Setup



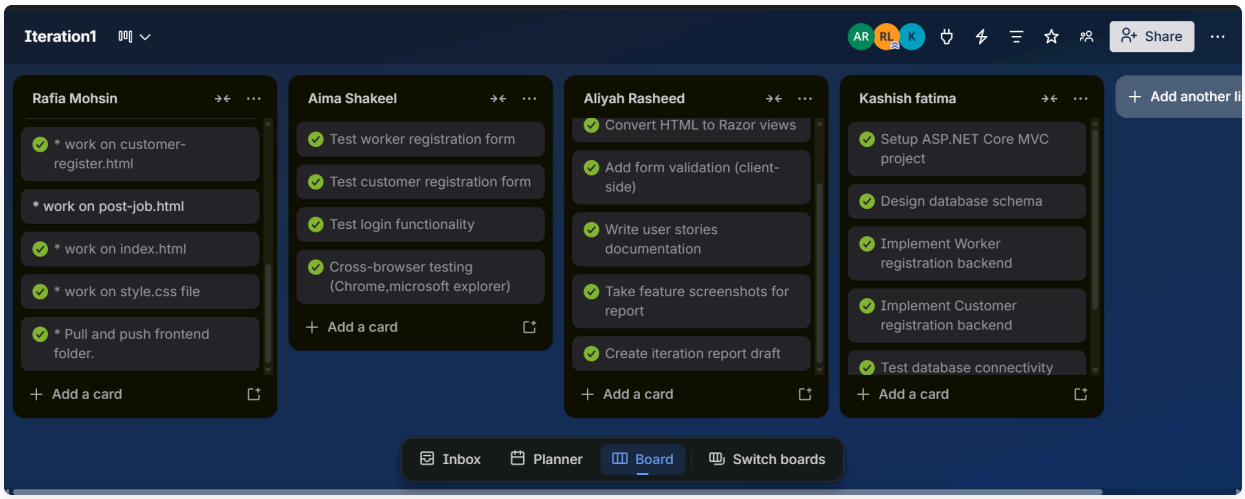
Initial board with tasks assigned to team members

Snapshot 2: Mid-Sprint Progress



Mid-iteration showing work in progress

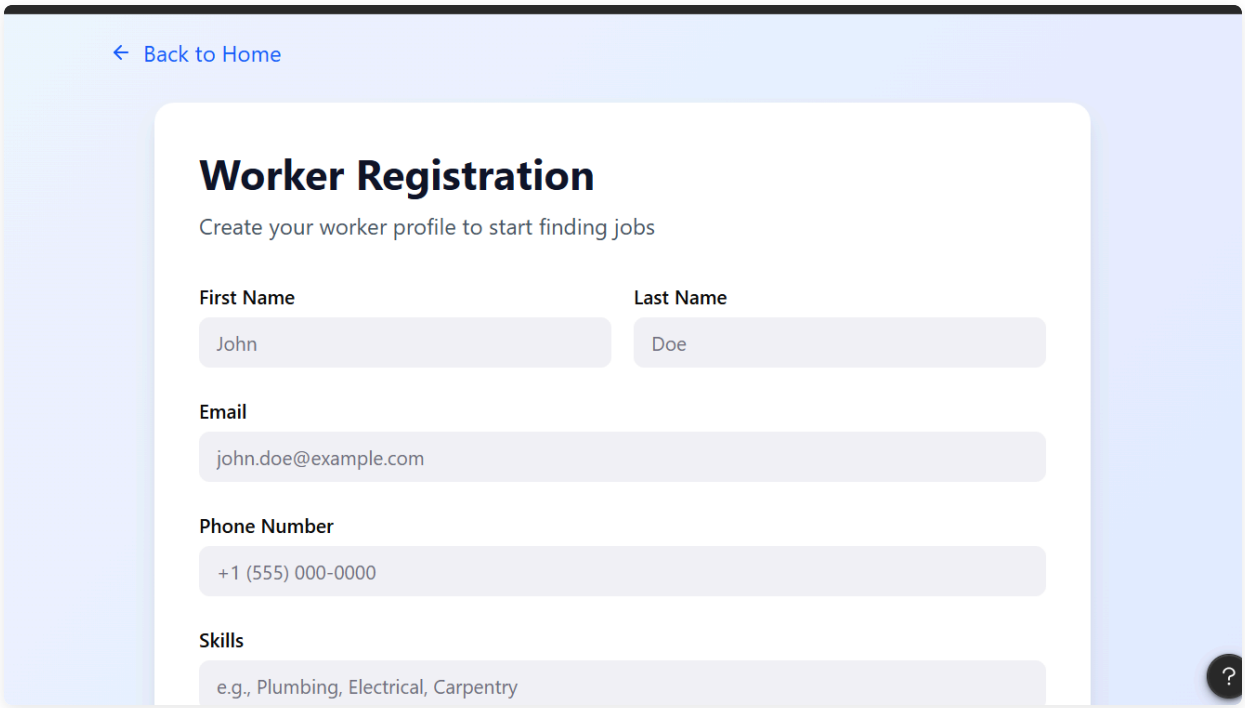
Snapshot 3: Final Board



Final state with all completed tasks marked ✓

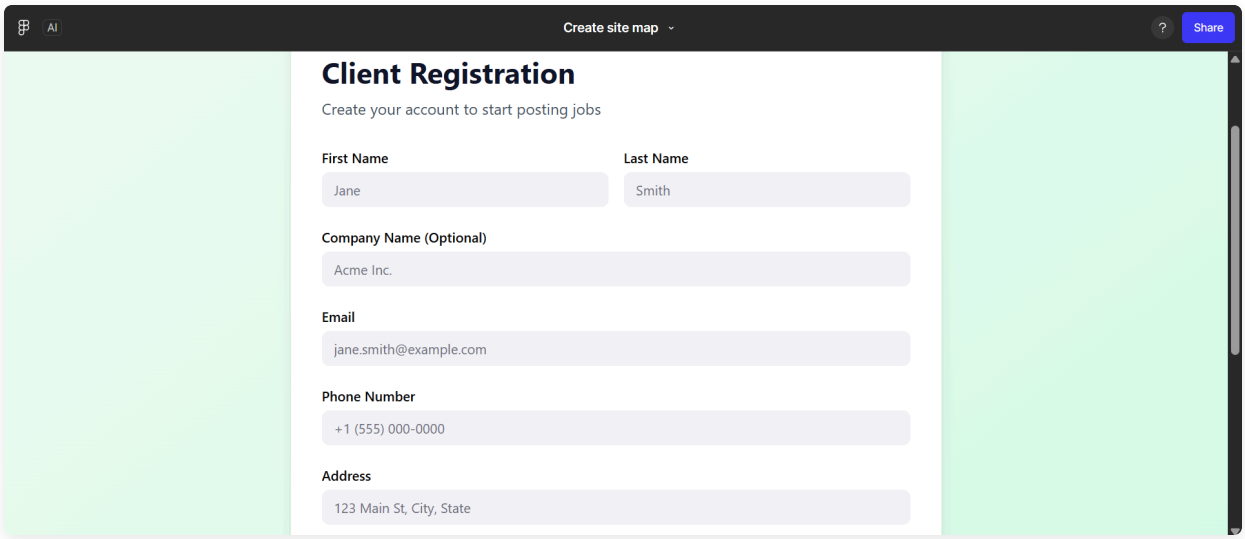
8. Feature Screenshots

Worker Registration Page



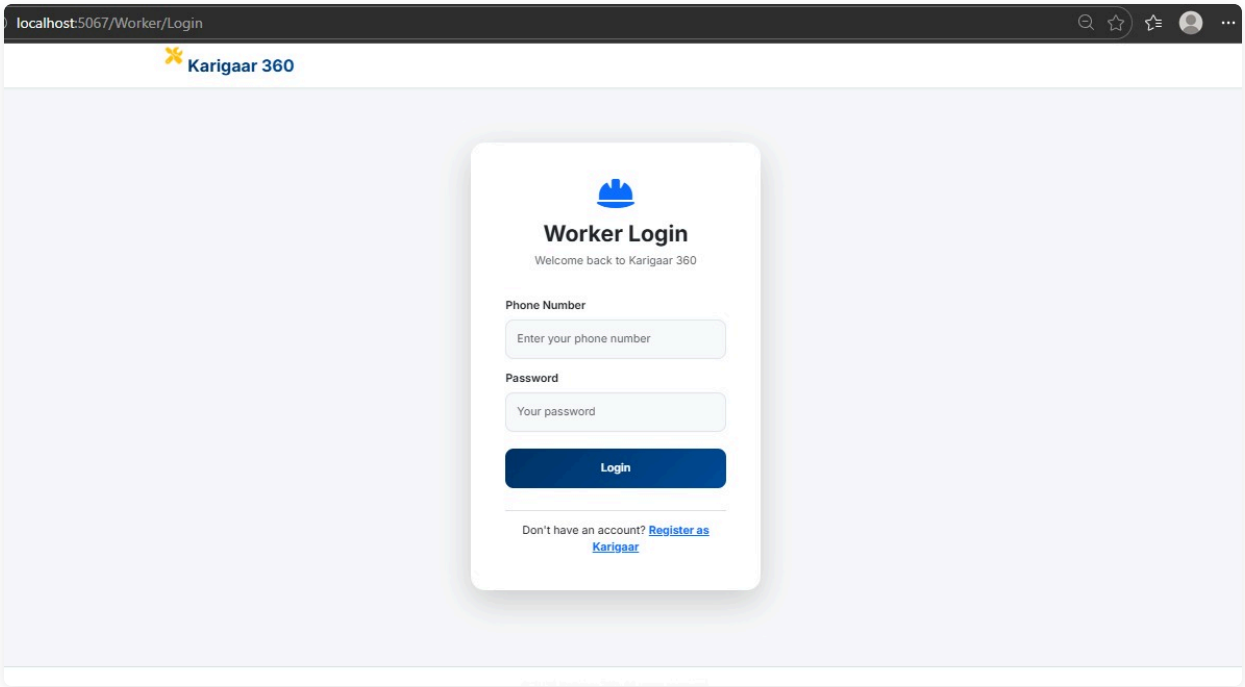
Workers can register with their professional details

Customer Registration Page



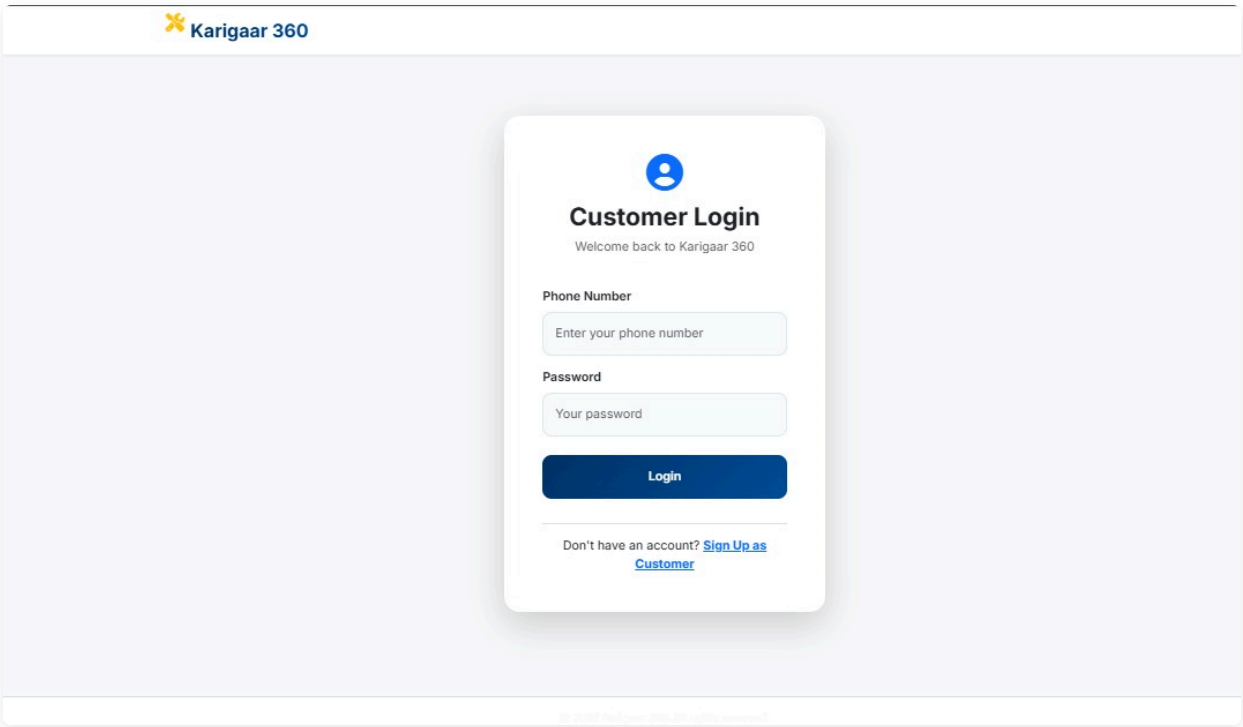
Customers can create accounts to post jobs

Worker Login Page



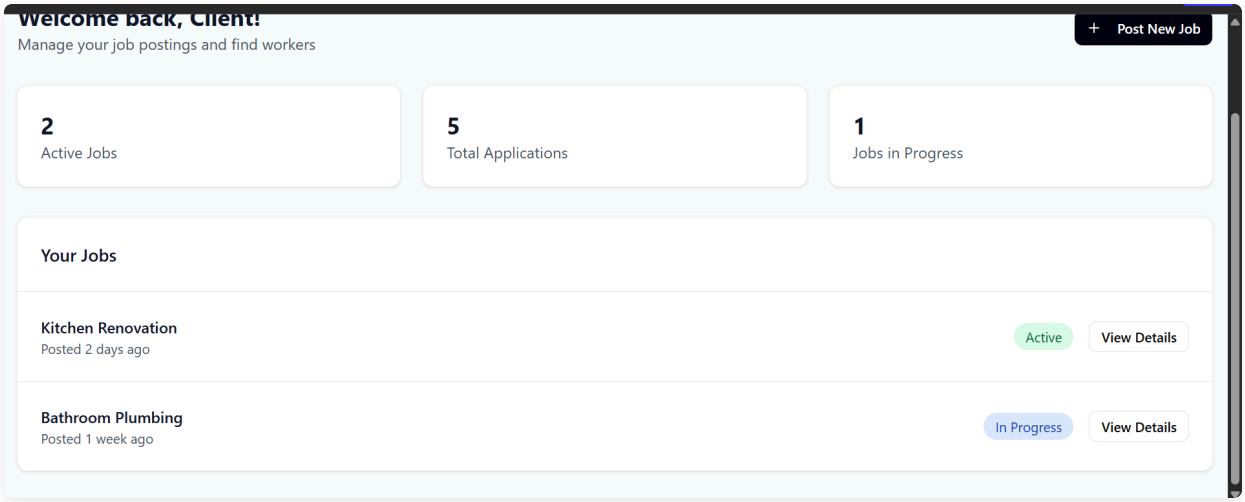
Workers can log in with phone number and password

Customer Login Page



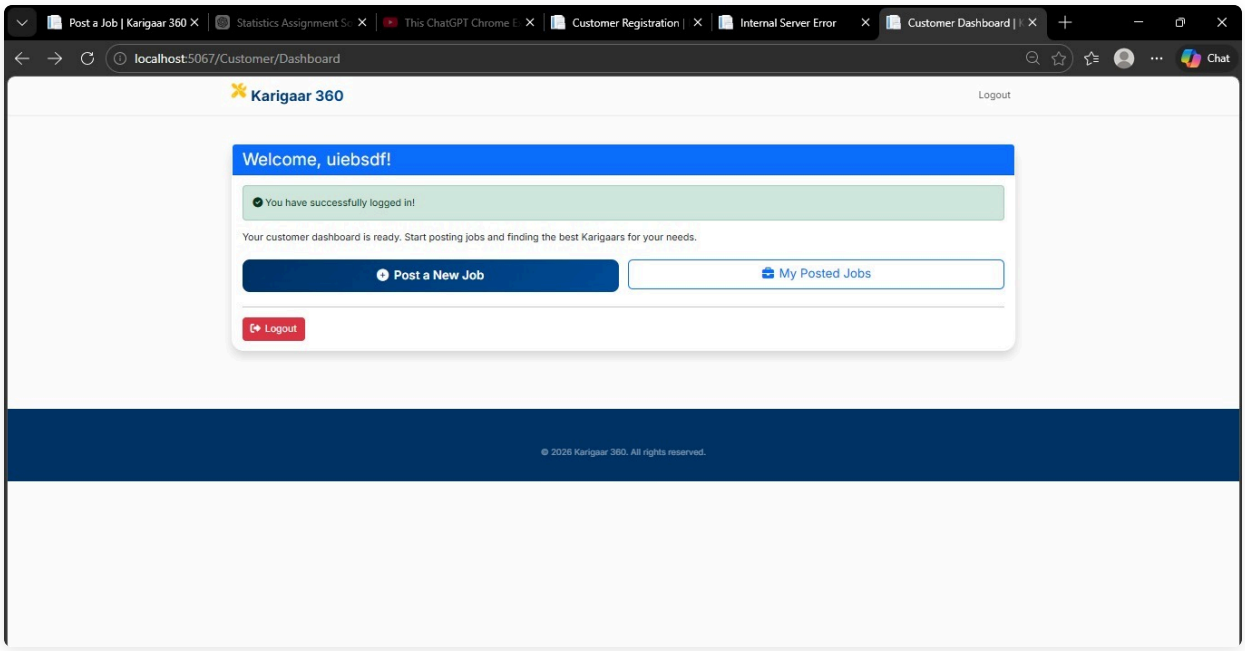
Customers can log in to access their dashboard

Worker Dashboard



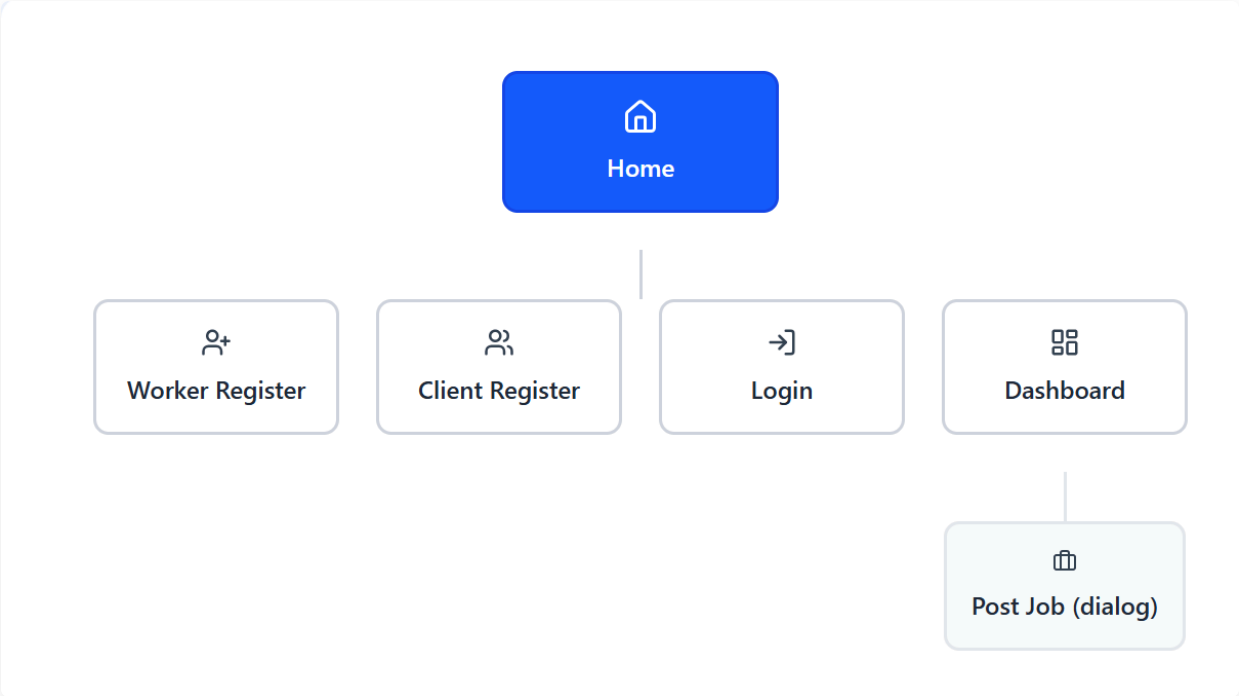
Worker dashboard after successful login

Customer Dashboard



Customer dashboard with job posting options

9. Figma Site Map



Site architecture planned using Figma showing user flows: Home → Worker Register, Client Register, Login, Dashboard, Post Job

Link: https://www.figma.com/make/FdDniVa68M7jm1rKmdaAw/Create-site-map

10. Challenges & Solutions

Challenge	Impact	Solution
Database connection issues	2-hour delay	Resolved connection string and SQL Server authentication
Password hashing implementation	1-hour delay	Used BCrypt.Net-Next with proper work factor
CSS path issues in Razor views	Minor styling break	Moved CSS to wwwroot, updated references
Phone number validation format	User input errors	Added regex pattern with clear error messages
Edge cases not handled	Backlog	Documented for Iteration 2

11. What Worked Well

✓ Parallel Development – Frontend and backend work overlapped efficiently
✓ GitHub Projects – Clear task tracking with team member columns
✓ Bootstrap Framework – Accelerated responsive design
✓ Entity Framework Core – Simplified database operations
✓ Regular Communication – Daily check-ins kept team aligned
✓ Figma Site Map – Clear visualization of user flows

12. Lessons Learned

- **Start backend earlier** – Could have overlapped more with frontend development
- **Document edge cases** – Some validation scenarios were missed initially
- **Create test data scripts** – Speeds up manual testing significantly
- **Add more comprehensive validation** – Will implement in Iteration 2

13. Backlog for Iteration 2

Feature	Priority	Est. Hours	Assigned To
Edge case validation handling	High	4	Aima
Job posting backend	High	6	Kashish
Worker dashboard enhancements	High	5	Aliyah
Customer dashboard enhancements	High	5	Aliyah
Job listing/search functionality	Medium	4	Aima
Rating and review system	Medium	4	Rafia
Unit testing implementation	High	4	Team
Total		**32 hours**	

14. Appendix

A. GitHub Repository Structure

```
Karigaar-360-Blue-Collar-Workforce-Platform/
├── backend/
│   ├── Controllers/
│   │   ├── HomeController.cs
│   │   ├── WorkerController.cs
│   │   └── CustomerController.cs
│   ├── Models/
│   │   ├── Worker.cs
│   │   └── Customer.cs
│   ├── Views/
│   │   ├── Home/
│   │   ├── Worker/
│   │   └── Customer/
│   ├── Data/
│   │   └── ApplicationDbContext.cs
│   └── wwwroot/
│       └── css/
│           └── style.css
├── frontend/
│   └── css/
├── docs/
│   └── Iteration_1_Report.pdf
└── README.md
```

B. Technology Stack

Backend	ASP.NET Core MVC 8.0
Frontend	Bootstrap 5.3, Font Awesome 6, Custom CSS
Database	Microsoft SQL Server 2022
ORM	Entity Framework Core 8.0
Security	BCrypt.Net-Next

Version Control	Git, GitHub
Project Management	GitHub Projects
Design	Figma

C. Links

- **GitHub Repository:** https://github.com/RafiaMohsin/Karigaar-360-Blue-Collar-Workforce-Platform
- **Figma Site Map:** https://www.figma.com/make/FdDniVa68M7jm1rKmdaAw/Create-site-map

15. Signatures

Role	Name	Registration ID	Date
Frontend Lead	Aliyah Rasheed	24L-2544	March 24, 2026
Backend Lead	Aima Shakeel	24L-2531	March 24, 2026
Backend Developer	Kashish Fatima	24L-2605	March 24, 2026
Project Coordinator	Rafia Mohsin	24L-2595	March 24, 2026

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